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Research paper

Efficiency of the EU regulation on greenhouse gas emissions in Italy: The hierarchical cluster analysis approach

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ABSTRACT

Using both a multivariate technique and a classic convergence analysis, we set out to test in an original way if the European Union (EU) regulation on greenhouse gas (GHG) emissions produced by urban road transport has generated a policy transfer process. By doing so, this paper attempts to fill the gap in research on ex-post analysis to assess whether the decisions taken at the EU level are actually transposed and implemented locally. The results obtained allow to document whether the EU regulation has stimulated the adoption of sustainable mobility policies for a selected group of Italian cities. The findings show that the processes of policy transfer may produce different effects if heterogeneity exists between cities or regions, even if they belong to the same country. Italian evidence demonstrates that northern cities are more actively drawing policy lessons. The study also reveals that large cities choose to look abroad more often than small municipalities. These results indicate that there may be institutional and economic variables that influence the transfer process.

1. Introduction

The vastly growing literature highlights that mitigating global climate change and neutralizing the impacts of fossil fuel-based energy policy on the environment have emerged as the biggest challenges for the planet, threatening both natural and built systems with long-term consequences (Yigitcanlar and Dizdaroglu, 2015). Mobility-related activities are deemed to play a major role in generating externalities for both of the natural and built environments (Dizdaroglu and Yigitcanlar, 2016, 2014). Consequently, sustainability of transport activities, as part of the sustainable urban development movement, has become a priority policy issue for many cities and countries across the globe (Ioppolo et al., 2014; Kamruzzaman et al., 2015, 2014).

Concerns related to the sustainability of transport activities suggest new policy models (Yigitcanlar and Kamruzzaman, 2014), which can also be identified through the process of learning from previous experiences in other historical and/or geographical contexts. The concept of adoption of political choices elaborated outside of local borders, so-called the lesson drawing or policy transfer process, remains particularly underdeveloped in the field of transport (Macmillen and Stead, 2014).

The scientific debate seems to pay little attention to the process of policy transfer in this field. Marsden and Stead (2011), through a systematic review of international literature, found only seven research papers that address this issue from the year 2000 to 2010. In this regard, van den Bergh et al. (2007) emphasize that the research that focuses on the issue of sustainability of transport activity examines exclusively the ex-ante of policies and the hypothetical socioeconomic consequences, almost completely ignoring the successes and failures actually recorded in this area.

Observing a selected group of Italian municipalities, the objective of this research is to verify the existence and the effectiveness of a policy transfer process generated by the European Union (EU) policies concerning the greenhouse gas (GHG) emission reduction. By analyzing data on the environmental sustainability of urban transport measured in Italian cities with a population of over 100,000 in 2006 and 2013, the study seeks to obtain the following outcomes. Firstly, we want to reach the goal to verify whether, in the local context examined, there has been a policy transfer regarding the transport policy defined at the EU level. Secondly, assuming that there may have been some processes of policy transfer, we want to verify the level of efficiency in terms of the EU policy target—i.e. the reduction of the environmental impact of urban transport. These two fundamental issues relating to the existence and effectiveness of policy transfer require ad hoc analysis. In order to

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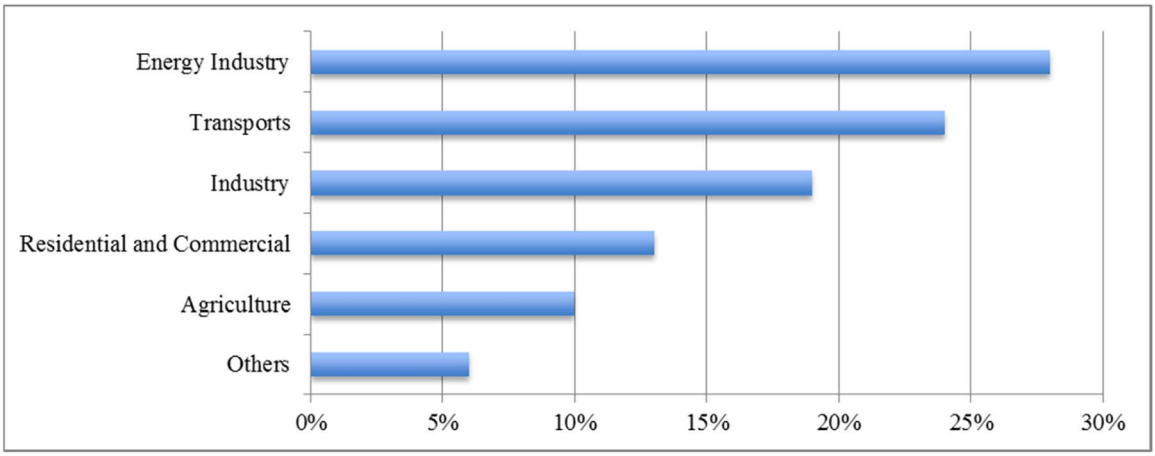

Fig. 1. GHG emissions by industries and households, EU 28, 2015.
Data from Euromobility (2016) and Eurostat (2014).

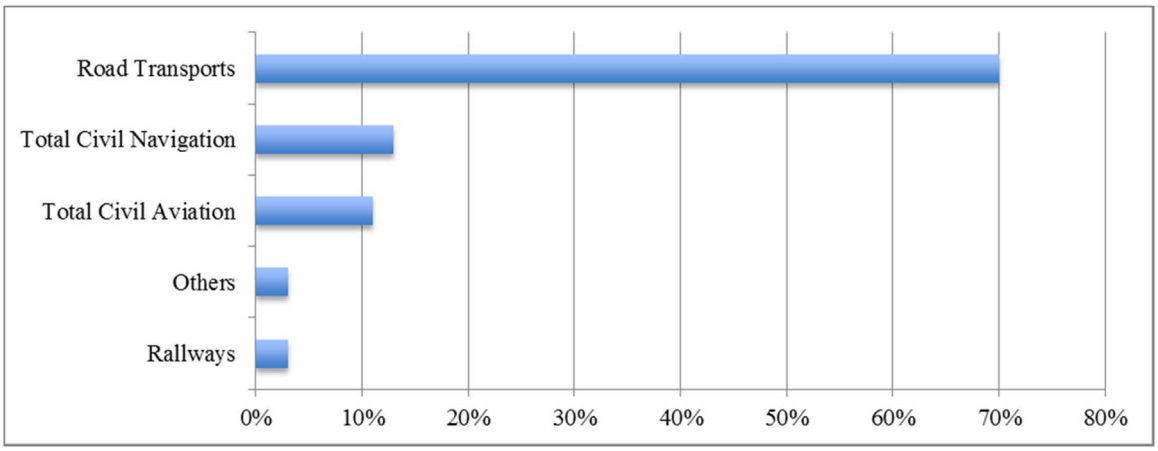

Fig. 2. GHG emissions by transport mode, EU 28, 2015.
Data from Euromobility (2016) and Eurostat (2014).

Table 1
Main policy measures included in the Transport White Paper.

Legislative act	Description
Fuel Quality Directive (*Directive 2009/30/EC*)	Greenhouse gas intensity in the energy used in road and non-road transport
Regulation on CO_2 from cars (*Regulation 443/2009*)	Emission performance standards for new passenger cars
Labeling for tires (*Regulation 1222/2009*)	Labeling requirements to increase the safety and the environmental and economic efficiency of road transport
Euro VI for heavy duty vehicles (*Regulation 595/2009*)	Emissions—heavy duty vehicles (Euro VI)
Promotion of Clean and Energy Efficient Road Transport Vehicles (*Directive 2009/33/EC*)	Clean and energy efficient road transport vehicles

undertake such analysis, the evaluation will be structured in two integrated steps: i) Hierarchical Cluster Analysis; ii) Convergence Analysis.

The paper is structured as follows. The second section identifies the typologies of policy transfer and discusses several good examples of transport policy transfer. In the third section, we describe the EU regulations in the field of road transport sustainability. The fourth and fifth sections describe the methodology and provide relevant descriptive statistics, respectively. In the final section, we discuss the findings and provide policy suggestions aimed at encouraging policy makers to carry out ex-post evaluation of the effects of policy transfer.

2. Sustainability and transport policy transfer in the European Union

The processes of transfer of ideas, actions and policies in the transport sector have followed over the years in a variety of contexts and economic systems. Nevertheless, literature only marginally analyzes this approach. Moreover, official policy documents regarding programs and new policy do not explicitly refer to policy transfer practice. However, what emerges is that policy transfers are isolated events, i.e. not belonging to a structured development policy based on knowledge of the choices already made and results achieved. This is also due to the fact of the excessive fragmentation of the institutional environment in charge of planning, financing and operating transport activities (Carlucci et al., 2017b).

Following the definition given by Dolowitz and Marsh (1996), we

Table 2
Variables of the study.

1) Territorial variables	2) Private transport variables
Population density (inhabitants/sqm)	Gas powered cars (percentage)
Population (inhabitants)	LPG powered cars (percentage)
Car Motorization index (vehicles/100 inhabitants)	Methane powered cars (percentage)
Car density (vehicles/sqm)	Motorcycle motorization index (vehicles/100 inhabitants)
	Euro 0–2 cars (percentage)
	Euro 3 cars (percentage)
	Euro 4–6 cars (percentage)
	Euro 0 motorcycles (percentage)
	Euro 1–2 motorcycles (percentage)
	Euro 3 motorcycles (percentage)
	Car sharing – cars in fleet (number of cars)
	Car sharing – users (number of users)
	Car sharing – cars/pop. (cars/10.000 inhabitants)
	Car sharing – users/pop. (users/10.000 inhabitants)
	Bike sharing – bikes in fleet (number of bikes)
	Bike sharing – users (number of users)
	Bike sharing – bikes/pop. (bikes/10.000 inhabitants)
	Bike sharing – users/pop. (users/10.000 inhabitants)
3) Public transport supply variables	**4) Environmental variables**
Municipalities vehicles (vehicles/10.000 inhabitants)	PM10 – exceedance days (number of days)
Public Transp. – Supply (seats*km/inhabitants)	PM10 – annual average (micrograms/cbm)
Public Transp. – Demand (No. passengers/inhabitants)	NO2 – exceedance hours (number of hours)
Restricted Traffic Zones (sqm/inhabitants)	NO2 – annual average (micrograms/cbm)
Pedestrian Zone (sqm/inhabitants)	Index of accidents (accidents/1.000 inhabitants)
Bike lanes (km/10.000 inhabitants)	Index of mortality (deaths/100 accidents)
Toll Parking (No. stalls/1.000 vehicles)	
Park & Ride (No. stalls/1.000 vehicles)	

consider policy transfer as a process in which knowledge about policies, administrative arrangements, institutions and so on in one time and/or place is used in the development of policies, administrative arrangements and institutions in another time and/or place. Bulmer and Padgett (2005) identify different typologies of governance within the EU. Hierarchical governance, characterized by a coercive form of policy transfer, negotiated governance in which norms are transferred by negotiation, voluntary governance based on policy transfer facilitated by the EU institutions.

Marsden et al. (2011) observe that the implementation of policies already in use in other geographical areas may depend on a voluntary choice (horizontal transfer) or come from an indirect imposition, for example a financial one (vertical transfer), or from a direct imposition as in the case of market regulation.

European airline deregulation and deverticalization of the railways are examples of transport policy transfers. In the first case, a judgment of the Court of Justice (The Nouvelles Frontières judgment of April 1986) initiated a process of 'coercive' policy transfer, as the EU member states were obliged to adopt competition policy for air transport already in force in other economic sectors. The British Rail reorganization took place through a long process lasting from 1979 to 1997. This experience is worthy of note, not only because it is very innovative, but also it was inspired by reforms in sectors other than transport. British policy makers, in fact, did not undergo the influence of the prevailing models of railway restructuring around the world, but opted to transfer the experiences (lateral transfer) of the privatization of other public services at a national level, such as telecommunications, gas, and electricity, to the field of rail transport.

Started with the British rail privatization, the restructuring of European railways is another case of transport policy transfer. Indeed, the Directive 440/91 imposed on the EU member states the separation of the network from the service, at least from an accounting perspective, in order to outline an evolutionary path for the European rail market, providing for competition among several operators. The reorganization of the German railways was based on careful analysis of the experiences gained in other countries, such as Sweden, Japan, Austria, Switzerland and the Netherlands, where the picture emerging from the situation in the EU was used only to strengthen the political consensus of the reform (Lodge, 2003).

From an institutional point of view, the policies transferred within a multinational organization, such as the EU, are agreed by the member states and thus the transfer process can be defined as negotiated transfer (Bulmer and Padgett, 2005). For example, in the urban mobility sector the EU fulfills the role of encouraging wide public consultation and coordination among member states in order to implement a negotiated policy transfer. Moreover, financial instruments, such as the Structural Funds and the Cohesion Fund, can be used to improve negotiation skills and achieve the goal of exchange of good practice.

In order to improve efficiency in the transport sector it could be desirable to implement new transport policy models through the

Table 3
Printout on factor 1 by the active continuous variables.

Variable label	Coordinate	Weight	Mean	Standard deviation
Population density (inhabitants/sqm)	–0.79	50.00	1,823.46	1,763.40
Public Transp. - Supply (seats*km/inhabitants)	–0.76	50.00	3,556.47	2,401.50
Bike sharing – users	–0.76	19.00	3,462.79	4,467.60
Bike sharing - users/pop. (users/10.000 inhabitants)	–0.76	50.00	1,133.99	1,038.00
Public Transp. - Demand (No. passengers/inhabitants)	–0.70	48.00	135.39	140.80
Bike sharing - bikes in fleet (number of bikes)	–0.69	19.00	295.16	440.20
Bike sharing - bikes/pop. (bikes/10.000 inhabitants)	–0.64	16.00	6.98	3.70
Population (inhabitants)	–0.63	50.00	289,752.00	442,475.00
	MIDDLE AREA			
Bike lanes (km/10.000 inhabitants)	0.03	50.00	2.79	3.10
Gas powered cars (percentage)	0.04	50.00	8.33	5.00
Car Motorization index (vehicles/100 inhabitants)	0.06	50.00	65.37	23.60
Euro 0 motorcycles (percentage)	0.13	50.00	27.38	6.70
Euro 0 - 2 cars (percentage)	0.14	50.00	27.77	7.90
Methane powered cars (percentage)	0.15	50.00	2.70	2.90
Euro 3 cars (percentage)	0.36	50.00	17.91	3.20
Euro 1 - 2 motorcycles (percentage)	0.36	50.00	37.74	4.50

Table 4
Printout on factor 2 by the active continuous variables.

Variable label	Coordinate	Weight	Mean	Standard deviation
NO2 – annual average (micrograms/cbm)	0.16	48.00	2.63	5.84
Car density (vehicles/sqm)	0.19	50.00	1,133.99	1,038.00
Population density (inhabitants/sqm)	0.23	50.00	1,823.46	1,763.43
Car sharing – cars in fleet (number of cars)	0.27	11.00	52.27	20.19
Euro 3 cars (percentage)	0.34	50.00	17.91	3.17
Euro 1–2 motorcycles (percentage)	0.52	50.00	37.74	4.51
Euro 0–2 cars (percentage)	0.63	50.00	27.77	7.88
Middle Area				
Bike lanes (km/10.000 inhabitants)	−0.81	50.00	2,791.00	3.13
Gas powered cars (percentage)	−0.77	50.00	8,331.00	4.97
LPG powered cars (percentage)	−0.70	50.00	5.51	2.56
Methane powered cars (percentage)	−0.70	50.00	2.70	2.90
Euro 4–6 cars (percentage)	−0.60	50.00	54.25	10.21
Index of mortality (deaths/100 accidents)	−0.59	50.00	4.65	1.64
Index of accidents (accidents/1.000 inhabitants)	−0.59	50.00	4.65	1.64
Park & Ride (No. stalls/1.000 vehicles)	−0.45	50.00	22.40	24.67

mechanism of policy transfer. Public policy makers can identify suitable transport policy models to solve a specific problem by observing how a similar situation was addressed in the past or in other geographical contexts and transfer those experiences, with appropriate corrections, to the local context in which they operate.

As discussed above, in the literature many mechanisms of policy transfer are analysed. In this paper, we focus our analysis on the effects of negotiated policy transfer or negotiated governance, according to the definition made by Bulmer and Padgett (2005), referring to a process in which governments are driven by a supranational organism like the EU. Specifically, we deal with the effects of the EU regulation aimed at the reduction of GHG emissions produced by commercial vehicles (Márquez-Ramos, 2015). Air pollution remains a major problem across the industrialized countries and transport sector is one of the main sources of pollutant. The EU is also cutting down GHG emissions following specific behavioral guidelines. In this context, one of the objectives of the Europe 2020 strategy is to reduce GHG emissions by 20% compared to 1990 levels (Nocera et al., 2015). Moreover, 24% of the overall GHG emissions in the EU are generated by transport activities (Fig. 1). In this context, road transport accounts for 70% of the total emissions (Fig. 2). In order to reduce GHG emissions produced by road transport, the Commission's Transport White Paper includes several relevant policy measures adopted in 2009 (European Commission, 2012) (Table 1).

3. Methodology

In 2009, the EU approved several relevant policy measures inducing a rapid reduction in emissions (Table 1). Even if achieving emissions reductions is a continuous process, in this study, we select two periods along the time. The first one is 2006, before introduction of the abovementioned policy measures, and the second is 2013, four years after the EU policy where in the field of road transport emissions were consolidated.

We consider a sample of 50 Italian municipalities with more than 100,000 inhabitants and use data on the sustainable transport policy recorded over the period of 2006–2013 by the Osservatorio Permanente sulla Mobilità (*Observatory on sustainable mobility in Italy*) database (Euromobility, 2016).

The sustainability phenomenon is represented by 36 variables grouped into four classes (Table 2).

In the first group, 'Territorial variables', we collect all variables able to describe the characteristics of the observed locality. The second is formed of 'Private transport variables', bringing together all the transport variables of citizens, regardless of level of greenhouse gas emissions. The third group, 'Public transport supply variables', takes into account all the variables related to the supply structure of public transport services. Finally, the last group of variables is labeled as 'Environmental variables'. This group includes all variables linked to the environmental and community health.

In order to obtain more detailed information about the characteristics of each municipality in regards to all variables and periods considered, we first decided to perform a multidimensional analysis (MDA) by means of a Hierarchical Cluster Analysis based on a Principal Component Analysis (PCA) (Hotelling, 1933). The resulting Factorial Plan and Clusters, automatically generated by the Parti-Decla Procedure of the Decisia software Spad, turn out to be a good means of studying the dynamics of public policy and also particularly useful later for interpreting the convergence phenomenon between the considered observations. More in detail, a PCA allows us to interpret the relations between transport policies in the light of two latent factors resulting from the linear combinations of the original variables, while by means of the Cluster analysis we examine the level of similarity in the choices made by municipalities regarding to the sustainable transport policy. Municipalities showing similar behaviors are incorporated into the same cluster.

Then, in order to verify the impact of the policy on the territory, we calculate policy convergence using the main variables that should describe the results desired by the EU (Arbolino, 2011). Policy convergence thus describes the result of a process of policy change over time toward some common point (Abramovitz, 1986; Knill, 2005). In other words, policy convergence is an increase in the similarity between one or more characteristics of a certain policy and, thus, the diffusion of policies can be expected to result in an increase of policy homogeneity among municipalities. In this paper, in accordance with Young et al. (2008), which outline that sigma convergence may not be accompanied by beta convergence, we use the sigma convergence concept focusing on the degree of similarity increase, in regards to both distributions of multivariate analysis and all the cities considered.

The sigma-convergence describes the time trend of a considered variable, measured by standard deviation (Friedman, 1992; Quah, 1993; Barro and Sala-i-Martin, 2004). Following Sala-i-Martin's (1996) exposition, we assume that:

$$\sigma_{t+T} < \sigma_{\tau} \quad (1)$$

where σ_τ is the standard deviation of log $x(_{i,t})$ across municipalities i at the time t. In this way, we observe the time distribution.

In order to calculate the dispersion among the groups of municipalities considered—although they are compounds by statistical unity of different composition and they do not have any homogeneous average values—we used the coefficient of variation normalized for eliminating the distorting effects according to the following formula-

Fig. 3. Map of Italy and location of the cities under investigation.

tion:

$$CV = \frac{\sigma}{\sqrt{(n-1)\mu}} \tag{2}$$

where CV is the coefficient of variation, meaning the ratio of the standard deviation to the mean μ.

The examination of the environmental disparities trend is aimed at verifying whether, over time, there has been an increase (divergence) or decrease (convergence) in the standard deviation. Any reduction in dispersion in the mean pollution levels helps us to understand whether

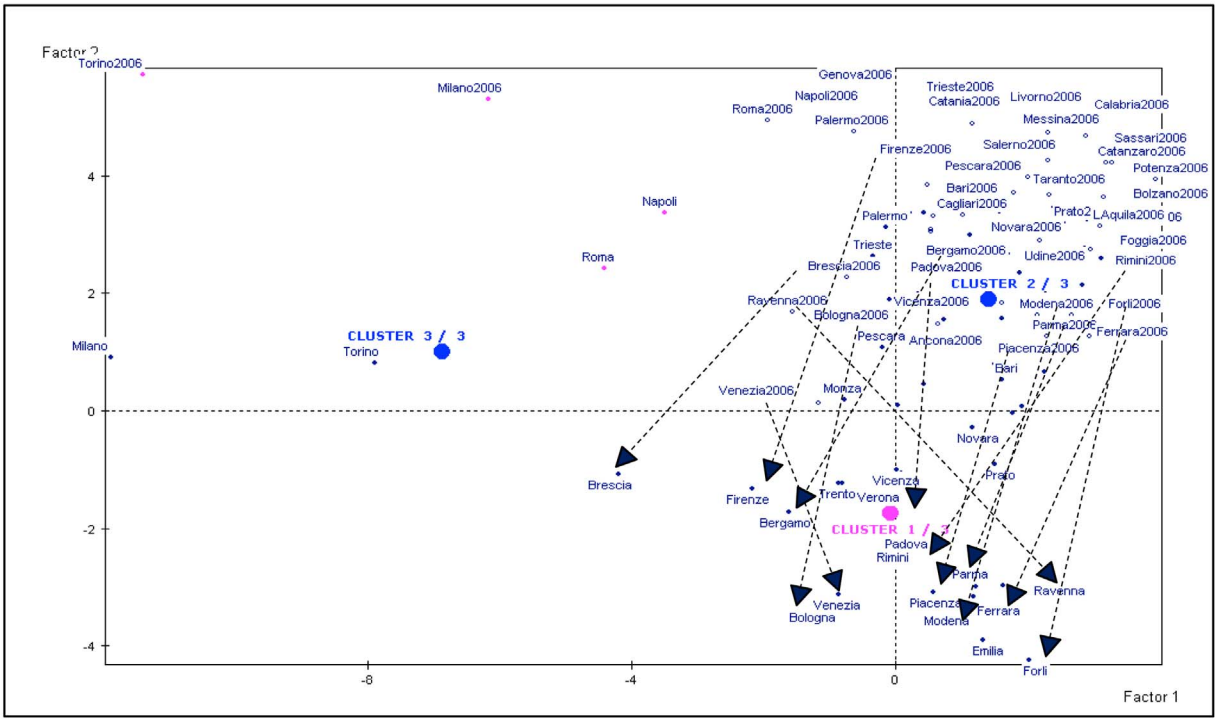

Fig. 4. Factorial plan with the projection of the variables.

Table 5
Cluster composition.

Clusters	1	2	3
	Piacenza	Naples	Latina
	Novara	Milan	Taranto
	Vicenza	Turin	R. Calabria
	Padua	Brescia	Messina
	Forli		Bari
	Verona		Catanzaro
	Trento		Syracuse
	R. Emilia		Sassari
	Prato		Pescara
	Udine		Campobasso
	Parma		Perugia
	Ferrara		Potenza
	Ancona		Aquila
	Ravenna		Cagliari
	Aosta		Catania
	Bolzano		Udine
	Bergamo		Salerno
	Florence		Trieste
	Genoa		Foggia
	Livorno		Palermo
	Monza		Terni
	Rimini		
	Rome		
	Modena		
	Bologna		
Inertia within clusters	0.80517	110.379	333.869
Inertia between clusters		402.125	

there has been progress regarding the sustainability of undeveloped areas.

4. The role of policy transfer in the Italian transport sector: a multivariate approach

In order to define the most suitable geographical area in which to observe the policy transfer process, we did not use geographical divisions established ex-ante, because it would not have been functional for our purpose. On the contrary, we aggregated the Italian territory using the methodology of group analysis through which it is possible to consider numerous important variables affecting the phenomenon examined (Flora and Arbolino, 2013).

Cluster analysis is an instrument able to evaluate several development factors at the same time. In this way, it was considered particularly suitable in order to evaluate the effectiveness of sustainable transport policy and to provide useful policy suggestions. In order to determine clusters of municipalities, it is important to identify an indicator to show performance or overall well-being for each statistical unit considered. The nature of this indicator is typically latent, i.e. it is a theoretical not directly observed construct underlying a series of manifest variables (MV) (Rashid and Yigitcanlar, 2015). Moreover, this indicator can be estimated only by observing MV (Fallowfield, 1996; Gertler, 1988).

By the means of PCA, we extracted from our dataset the first two latent factors, explaining 66.6% of the total variance, considering the year 2013. The first factor is positively characterized by variables that define the private transport mode (Table 3), while the second factor is characterized by the environmental transport policy (Table 4).

Fig. 4 illustrates the factorial plan with the projection of the variables considered. In particular, the first factor mainly includes the arrangements with individual consumption which cannot be ruled out. This arrangement shows operative and positive marginal costs (Nash, 2003). Even if overall, the detected latent variable is the private transport, we can observe a contrast between modalities of private transport in accordance with the principles of environmental sustainability and other modalities highly polluting (see Box 2 in Table 3). Moreover, the second latent factor is described by environmental variables; it can be noted that the value describing the vertical axis to the plan differs at the top for the low strength of GHG emissions, while at the bottom for a greater presence of the same values although they are represented by different indicators.

Using the Parti-Decla procedure of Spad, the software itself generates the number of clusters that maximizes the inter-cluster inertia and minimizes the intra-cluster inertia. The data processing detected three clusters that give an idea of the virtuosity environmental of the

Table 6
Cluster 1 characterization.

Characteristic variables	Cluster mean	Overall mean	Cluster std. deviation	Overall std. deviation	Test-value
Euro 4–6 cars (percentage)	60.86	54.25	8.86	10.21	4.53
Bike lanes (km/10.000 inhabitants)	4.63	2.79	3.33	3.13	4.1
Index of mortality (deaths/100 accidents)	5.57	4.65	1.56	1.64	3.92
Index of accidents (accidents/1.000 inhabitants)	5.57	4.65	1.56	1.64	3.92
Gas powered cars (percentage)	10.83	8.33	5.44	4.97	3.52
Methane powered cars (percentage)	4.04	2.70	3.39	2.90	3.24
LPG powered cars (percentage)	6.62	5.51	2.73	2.56	3.04
Euro 3 motorcycles (percentage)	36.81	34.65	6.27	5.95	2.54
Park & Ride (No. stalls/1.000 vehicles)	29.15	22.40	30.62	24.67	1.92
Toll Parking (No. stalls/1.000 vehicles)	55.52	47.10	39.71	33.14	1.78
Pedestrian Zone (sqm/inhabitants)	0.47	0.33	0.94	0.69	1.39
Restricted Traffic Zones (sqm/inhabitants)	7.93	7.46	8.57	9.59	1.34

Table 7
Cluster 2 characterization.

Characteristic variables	**Cluster mean**	**Overall mean**	**Cluster Std. deviation**	**Overall Std. deviation**	**Test- value**
Population density (inhabitants/sqm)	6,170.26	1,823.46	2,379.03	1,763.43	5.09
Car density (vehicles/sqm)	3,473.27	1,133.99	1,283.01	1,038.00	4.65
Car sharing - users/pop. (users/10.000 inhabitants)	19,294.00	3,462.79	5,783.91	7,247.46	4.01
PM10 - exceedance days (number of days)	102.50	44.32	20.62	31.44	3.83
NO2 - exceedance hours (number of hours)	13.25	2.63	10.55	5.96	3.68
PM10 - annual average (micrograms/cbm)	42.75	31.18	4.32	7.16	3.34
Public Transp. - Supply (seats*km/inhabitants)	7,326.79	3,556.47	4,214.78	2,401.49	3.24
Bike sharing - bikes/pop. (bikes/10.000 inhabitants)	17.97	6.98	6.95	6.53	3.13
Bike sharing - bikes in fleet (number of bikes)	1,464.00	295.16	1,257.85	714.02	3.01
Public Transp. - Demand (No. passengers/inhabitants)	315.13	135.39	217.39	143.71	2.59
Car sharing - users (number of users)	6,956.33	3,025.73	3,606.74	3,265.45	2.33
Car sharing - cars in fleet (number of cars)	81.67	52.27	53.54	43.04	1.32
Toll Parking (No. stalls/1.000 vehicles)	63.40	47.10	21.51	33.14	1.02

Table 8
Cluster 3 characterization.

Characteristic variables	Cluster mean	Overall mean	Cluster Std. deviation	Overall Std. deviation	Test-value
Euro 1–2 motorcycles (percentage)	41.02	37.74	3.90	4.51	4.32
Euro 0–2 cars (percentage)	33.12	27.77	4.63	7.88	4.04
Euro 3 cars (percentage)	19.96	17.91	1.18	3.17	3.86

municipalities considered (Fig. 3).

Cluster 1 (Tables 5 and 6) grouped the most virtuous cities. These cities share the presence of variables that represent the implementation of urban environmental sustainability policies and the absence of indicators defining the traditional transport model (involving many negative externalities). More specifically, it is characterized by the presence of EURO 4–6 cars (T value 4.53), bike lanes (T value 4.10), gas powered cars (T value 3.52) park & ride (T value 1.92), but also pedestrian zone, supply of public transport, limited traffic zones, that are mainly present in the Box 4 in Table 2. These variables exclusively characterize the cities of the center-north.

Cluster 2 (Tables 5 and 7) characterized by a high density of population and cars (T value 5.09), is reported among the territorial variables (see Box 1 in Table 3). In this cluster there are the metropolitan areas of Milan, Naples, Turin and Brescia; in addition to the usual transport supply, they offer new tools for sustainable private transport such as bike sharing (T value 4.01), and car sharing (T value 2.33) (see Box 2 in Table 3). This cluster is characterized by high levels of environmental pollution (days of annual exceeding − T value 3.83; hours of annual exceeding – T value 3.83) (see Box 4 in Table 3).

Cluster 3 (Tables 5 and 8) is characterized by a low level of sustainability. All variables considered present negative T values, except for Euro 1–2 motorbikes (T value 4.32) and Euro 0–2 cars (T value 4.04) and Euro 3 cars (T value 3.86). This cluster includes almost exclusively cities in the south, with the exception of Terni and Aquila (central Italy), and Udine and Trieste (northern Italy).

Returning to the factorial plan, we can evaluate the shifts experienced by cities over the whole period considered, with respect to all variables and to two latent variables, including the 2006 data for the cities in the analysis as additional observations. The arrows in Fig. 4 show the shifts experienced by the municipalities considered over the period examined. Thus, every municipality is shown twice: the first time, with the indication of 2006, suggests the situation in the absence of specific rules, the second (without value) defines the situation

Table 9
Sigma convergence.

Variables	Cluster1		Cluster 2		Cluster 3		Total	
	σ_t	σ_{t+T}	σ_t	σ_{t+T}	σ_t	σ_{t+T}	σ_t	σ_{t+T}
PM10 – exceedance days (number of days)	0.077	0.094	0.303	0.134	0.202	0.237	0.085	0.106
PM10 – annual average (micrograms/cbm)	0.038	0.036	0.183	0.067	0.065	0.058	0.037	0.033
NO2 – exceedance hours (number of hours)	0.622	0.453	0.6	0.531	0.42	0.603	0.38	0.323
NO2 – annual average (micrograms/cbm)	0.059	0.049	0.122	0.037	0.105	0.106	0.055	0.05

following the introduction of the policy. We observe that the policy transfer implementation from Cluster 3 (in rearmost position) to Cluster 1 only concerns cities in central and northern Italy. Arrows linking additional observations to active ones show the main shifts.

Finally, by verifying the impact of the policies, we calculated the convergence sigma indexes (Table 9). The variables used represent proxy to analyze environmental pollutants and can be considered as the result indicators of the policy. Overall, if we consider the national aggregate level (see the total column), the results of our estimates show that, between the two periods, targets predetermined by the EU policy were not achieved. Instead, some difference arises considering the disaggregated analysis. The data seem worrying with reference to Cluster 3, where all values increased; this underlines a divergence process. Instead, Cluster 2, despite being characterized by a high density of population and cars, shows the most tangible signs of a policy achieving good results.

5. Discussion and conclusion

The transplantation of policies already in use elsewhere has increased in the EU over the recent years, particularly in the fields of transport and urban development (Stead et al., 2008). Even if the experiences of policy transfer in the transport sector are certainly not an exception (Marsden et al., 2011), it is evident that, in most cases, policies do not involve the use of scientific methodologies.

For example, in railway sector, the process of policy transfer among the EU members started with the British rail privatization experience despite, as underlined by Bowman (2015), this experiment has been a failure.

Analyzing the diffusion process of the policies adopted in the transport area, one perceives a 'detachment' between reality and the level of interest in the scientific community. In the EU, the academic influence on transport policy makers is still limited compared with the North America (Bray et al., 2011). Moreover, there is a lack of logical analysis and conceptual aims to provide policy makers with a tool that can support them in their choice of strategies.

In order to contribute to bridging the gap, this study explores the role of a scientific methodology for evaluating the effectiveness of a negotiated policy transfer process in the Italian urban transport context. The reveals insights and the findings should help policy makers to identify good practices and define new policy instruments. Focusing on a selected group of Italian cities, the analysis verifies whether a policy transfer has occurred from the EU and assesses the effectiveness of this process in terms of reduction of CO_2 emissions.

We carried out this work by using hierarchical cluster analysis and convergence analysis. The rationale behind a statistical approach (Karlaftis and Vlahogianni, 2011) was to better analyze the EU transport policies and consequences at local level.

We considered cluster analysis particularly suitable in order to evaluate the level of similarity in the choices made by municipalities regarding sustainable transport policy, i.e. to verify a process of policy transfer. Moreover, in order to evaluate the impact on the territory of the policy transfer process, we calculated policy convergence using the main variables that should describe the EU policy targets.

The results recall the historical imbalance in the growth rate between the northern and southern Italian cities (Carlucci et al., 2017a), with the implementation of policies for environmental sustainability being in the northern cities more incisive and dynamic, while the southern municipalities show difficulties in aligning themselves to European Directives. In particular, our results paint a picture in which the cities of the northern Italy show a progressive transition from the old public and private transport system to a new one, focused on tools aimed at pollution reduction, while this process is absent in the southern Italy.

A separate group is formed of certain metropolitan cities, including Naples, Milan, Brescia and Turin, which have always been characterized by a strong commitment to local public transport policies. Once again, our analysis confirms this tendency and clearly highlights the differences between these large cities and cities of smaller size. Finally, our analysis highlights that metropolitan areas have achieved better performances than the smaller municipalities in terms of reaching the environmental targets laid down by the EU directives.

The main insight of the analysis is that there can be different degrees of negotiated policy transfer whenever heterogeneity exists among cities, even if they belong to the same country. Italian evidence shows that northern cities are more actively drawing policy lessons. Another insight is also generated with regards to the circumstance that larger cities choose to look abroad more often for best practices than smaller municipalities. These results document that there may be institutional and economic variables that influence the transfer process.

Furthermore, the previous discussion underlines another possible caveat in our analysis, i.e., it may still be necessary to analyze many other considerations in order to exchange of policy between the EU member states. In this regard, Fink (2013) suggested that the EU does not enhances policy learning among member countries, even if the EU is considered as an 'environment' that facilitates policy transfer.

From these considerations, it appears to be that policy makers should carefully evaluate the potential of techniques for designing policies based on the analysis of strategies adopted elsewhere and their successes and failures. At the same time, the scientific world should investigate this issue with a greater level of detail, particularly in the transport sector and sustainable mobility. In this field much research has focused on investigating transport emissions and their impacts (François et al., 2017) and on the creation and the use of appropriate indicators to measure the sustainability of transport systems (Alonso et al., 2015; Gudmundsson and Sørensen, 2013; Reisi et al., 2014), whereas studies involving the transfer of technical policies are an exception.

Even at a local level, where theoretically the coordination of mobility should be more intense, in reality this does not occur. One possible solution to this problem is to build a new form of public-private transport-oriented environmental governance focusing on sustainable urban development concerns (Ioppolo et al., 2016).

Currently, maybe due to the prevalence of engineering-style approaches to the governance of transport systems and mobility, particularly in Italy, policy transfer practice is still at a negligible level. The analysis reported in this paper underlines the untapped potential of policy transfer as a tool for designing appropriate policies to ensure an improvement in the organization of the transport market. Realizing such potential will assist in contributing to the solutions to global challenges, and moving towards a more sustainable urban future and generating common prosperity (Yigitcanlar et al., 2015).

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